

Artificial Intelligence Homework 2

April 16, 2026

Q1: AC-3 puts back on the queue every arc (X_k, X_i) whenever any value is deleted from the domain of X_i , even if each value of X_k is consistent with several remaining values of X_i . Suppose that, for every arc (X_k, X_i) , we keep track of the number of remaining values of X_i that are consistent with each value of X_k . Explain how to update these numbers efficiently and hence show that arc consistency can be enforced in total time $O(n^2d^2)$.

Q2: Consider the problem of placing k knights on an $n \times n$ chessboard such that no two knights are attacking each other, where k is given and $k \leq n^2$.

- Choose a CSP formulation. In your formulation, what are the variables?
- What are the possible values of each variable?
- What sets of variables are constrained, and how?
- Now consider the problem of putting as many knights as possible on the board without any attacks. Explain how to solve this with local search by defining appropriate ACTIONS and RESULT functions and a sensible objective function.

Q3: Consider the following sentence:

$$[(Food \Rightarrow Party) \vee (Drinks \Rightarrow Party)] \Rightarrow [(Food \wedge Drinks) \Rightarrow Party].$$

- Determine, using enumeration, whether this sentence is valid, satisfiable (but not valid), or unsatisfiable.
- Convert the left-hand and right-hand sides of the main implication into CNF, showing each step, and explain how the results confirm your answer to (a).

c. Prove your answer to (a) using resolution.

Q4: Prove each of the following assertions:

- (a) α is valid if and only if $True \models \alpha$.
- (b) For any α , $False \models \alpha$.
- (c) $\alpha \models \beta$ if and only if the sentence $(\alpha \Rightarrow \beta)$ is valid.
- (d) $\alpha \equiv \beta$ if and only if the sentence $(\alpha \Leftrightarrow \beta)$ is valid.
- (e) $\alpha \models \beta$ if and only if the sentence $(\alpha \wedge \neg\beta)$ is unsatisfiable.